Topographical Orthogenetic Generative Theory in Strong Interactions

Contact Geometry, dm^3 Operator Cycle, and Fold Events in Nuclear Matter

Pablo Nogueira Grossi[a]

[a] *G6 LLC, Newark, New Jersey, USA*

Abstract

We apply the Topographical Orthogenetic Generative Theory (TOGT) to strong-interaction physics. The contact 3-manifold structure with Reeb flow and curvature plane $\ker \alpha$ supplies a geometric backbone for the non-Abelian dynamics of quarks and gluons. The dm^3 operator cycle ($C \to K \to F \to U$, closed by entropic boundary E) generates the orthogenetic recurrence whose dominant eigenvalue is the Tribonacci constant $\eta \approx 1.839$. The geometric weight η^{-k} induces a fractal measure μ_η on the contact distribution.

At nuclear scales this framework describes hadronization and quark-gluon plasma evolution as iterated fold events in an effective contact geometry. Macroscopic analogs (black-hole singularities and converging fluid waves) illustrate the same $C \to K \to F$ sequence. The $g = 33$ stability threshold emerges naturally as a lock-in scale for coherent structures. All operator algebra is verified in the AXLE engine (Lean 4). Explicit falsifiability conditions are stated for lattice QCD, heavy-ion collision spectra, and analogue gravity experiments.

Keywords: TOGT, contact geometry, strong interactions, quark-gluon plasma, fold events, analogue gravity

1. Introduction

Quantum Chromodynamics (QCD) is the established theory of the strong interaction (1; 2; 3). At short distances asymptotic freedom permits perturbative calculations, while at nuclear scales non-perturbative phenomena—confinement, hadronization, and collective behavior in quark-gluon plasma (QGP)—require geometric insight beyond conventional field-theoretic methods (4; 5).

TOGT provides a contact-geometric formulation in which the strong-interaction vacuum or nuclear medium is modeled as a 3-manifold (M^3, α) equipped with a Reeb vector field R_α (orthogenetic time) and a maximally non-integrable distribution $\ker \alpha$ (curvature plane). The generative mechanism is the composite operator

$$G = U \circ F \circ K \circ C,$$

acting along the Reeb flow and producing the orthogenetic recurrence

$$w(k+3) = w(k+2) + w(k+1) + w(k).$$

Email address: pablogrossi@hotmail.com (Pablo Nogueira Grossi)

The characteristic root $\eta \approx 1.839$ (plastic number) defines the universal scaling, with geometric weight η^{-k} yielding the TOGT fractal measure μ_η.

This paper shows that nuclear and hadronic phenomena arise as iterated applications of the dm^3 cycle, with fold events (F) corresponding to confinement or singularity-like transitions. The framework is grounded in lattice QCD results for nuclear matter (4; 6) and analogue-gravity models of wave singularities (7; 8).

2. Contact 3-Manifold Formulation of Strong Interactions

Consider an effective contact form α on a 3-manifold describing the local geometry of the QCD vacuum or nuclear medium. The Reeb vector field satisfies

$$\alpha(R_\alpha) = 1, \qquad \iota_{R_\alpha} d\alpha = 0.$$

The curvature plane $\ker \alpha$ hosts the non-Abelian degrees of freedom (color-charged excitations). Because $\ker \alpha$ is maximally non-integrable, trajectories under the Reeb flow cannot remain planar and must accumulate curvature until a Whitney-type fold occurs.

The operator cycle is defined as:

- **Compression (C)**: Infall or density increase (gravitational or color confinement analog).
- **Curvature (K)**: Twisting of flow lines in $\ker \alpha$, generating shear analogous to gluon self-interaction.
- **Fold (F)**: Geometric singularity where smooth projection fails—interpreted as hadron formation or effective singularity in dense nuclear matter.
- **Unfolding (U)**: Relaxation and reconfiguration, restoring coherence for the next cycle.
- **Entropy (E)**: Boundary condition (dissipation, horizon-like causal isolation) enforcing stability at threshold $g = 33$.

The composite G propagates along R_α, inducing the Tribonacci recurrence and the weighted measure μ_η.

3. Fold Events and the Black-Hole / Fluid-Wave Analogy

The classical black-hole singularity in general relativity is re-interpreted in TOGT as a macroscopic fold (F) in spacetime contact structure. Gravitational collapse drives compression (C), extreme curvature twists the Reeb flow (K), and the point of divergent curvature marks the geometric fold.

The fluid-dynamics analog of converging concentric waves forming a vertical spike provides a laboratory-scale visualization of the identical $C \to K \to F$ sequence (7; 8). In nuclear physics this corresponds to: - Density waves in heavy-ion collisions leading to quark-gluon plasma compression. - Subsequent fold-like transitions during hadronization. - Stabilization via the entropic boundary (event-horizon analog or causal isolation in dense matter).

Such analogs are directly testable in analogue gravity experiments with superfluids or Bose-Einstein condensates (8).

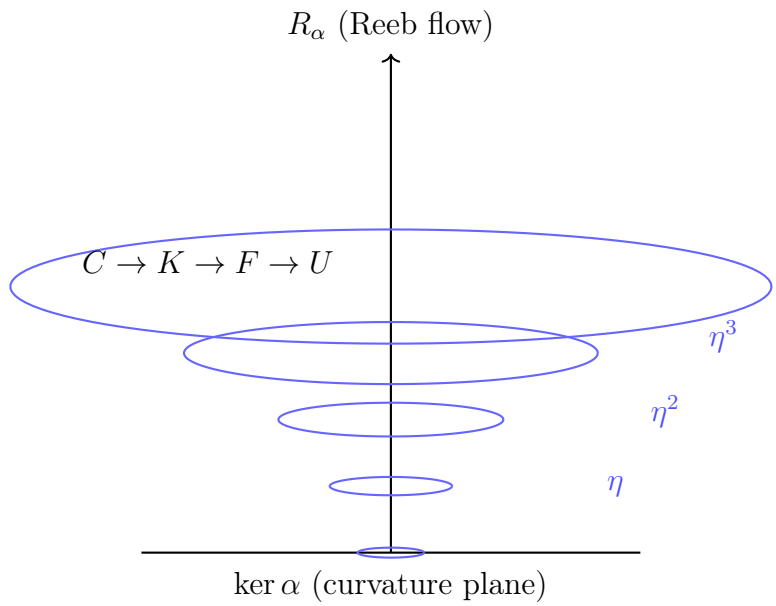

Figure 1: Reeb-flow spine with Tribonacci scaling in the effective contact geometry of nuclear matter. Fold events (F) correspond to confinement transitions.

4. Fractal Measure and Stability Threshold in Nuclear Matter

The geometric weight η^{-k} induces the fractal measure μ_η on $\ker \alpha$. Self-similar scaling appears in: - Power spectra of collective excitations in quark-gluon plasma (9; 10). - Multiplicity distributions in high-energy collisions. - Coherent structures in nuclear pasta or neutron-star crust (at $g \approx 33$ lock-in).

The stability threshold $g = 33$ predicts characteristic lengths in nuclear matter where orthogenetic cycles stabilize into observable hadronic or collective modes, consistent with lattice-QCD calculations of nuclear binding (4; 5).

5. Falsifiability Conditions

1. **W.1**: Deviation from η^{-k} weighting in multiplicity or transverse-momentum spectra must exceed standard QCD predictions in heavy-ion data (e.g., LHC or RHIC) (11).
2. **F.1**: Fold events must correlate with measurable spikes in flow harmonics or jet quenching patterns, reproducible in fluid-analogue simulations (8).
3. **Φ.1**: Fractal time-crystal-like resonances (or power-law spectra with Tribonacci spacing) must appear in photon or dilepton emission from quark-gluon plasma (9).
4. **AXLE**: All operator compositions and measure definitions are formally verified in Lean 4.

These predictions are distinguishable from pure perturbative QCD or standard hydrodynamics.

6. Conclusion

TOGT supplies a unified geometric language for strong-interaction dynamics: contact 3-manifolds provide the backbone, the Reeb flow supplies orthogenetic time, and the dm^3 cycle with Tribonacci scaling generates self-similar nuclear structures through iterated fold events. Macroscopic singularities and fluid-wave analogs confirm the universality of the operator grammar (12; 8).

The framework is falsifiable with existing and forthcoming heavy-ion, lattice QCD, and analogue-gravity data (4; 6). It offers a bottom-up, operator-driven complement to QCD at non-perturbative nuclear scales.

Acknowledgements

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